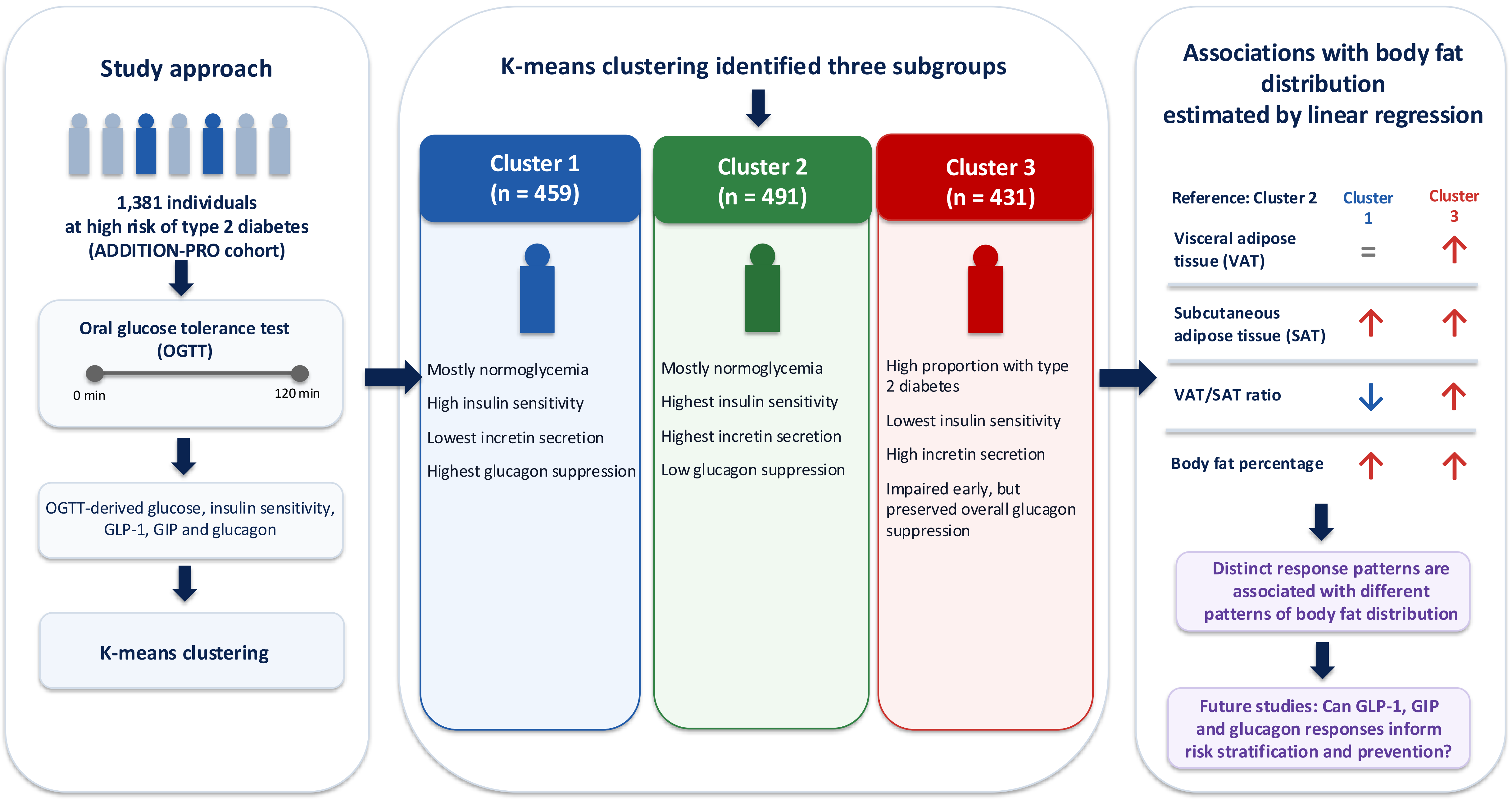
Study approach
1,381 individuals
at high risk of type 2 diabetes
(ADDITION-PRO cohort)
Oral glucose tolerance test
(OGTT)
0 min
120 min
OGTT-derived glucose, insulin sensitivity,
GLP-1, GIP and glucagon
K-means clustering
K-means clustering identified three subgroups
Cluster 1
(n = 459)
Mostly normoglycemia
High insulin sensitivity
Lowest incretin secretion
Highest glucagon suppression
Cluster 2
(n = 491)
Mostly normoglycemia
Highest insulin sensitivity
Highest incretin secretion
Low glucagon suppression
Cluster 3
(n = 431)
High proportion with type
2 diabetes
Lowest insulin sensitivity
High incretin secretion
Impaired early, but
preserved overall glucagon
suppression
Associations with body fat
distribution
estimated by linear regression
Reference: Cluster 2
Cluster
1
Cluster
3
Visceral adipose
tissue (VAT)
=
Subcutaneous
adipose tissue (SAT)
VAT/SAT ratio
Body fat percentage
Distinct response patterns are
associated with different
patterns of body fat distribution
Future studies: Can GLP-1, GIP
and glucagon responses inform
risk stratification and prevention?